HYUN DONG LEE

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Research focus: compression of context, reasoning, and knowledge in language models through new architectures, algorithms, and synthetic data

EDUCATION

Stanford University, School of Engineering **Stanford, CA**
Ph.D. in Computer Science (GPA: 4.03/4.30) Sep. 2022 - Present
- Advisors: Prof. Scott Linderman, Prof. Emily Fox

Columbia University, School of Engineering and Applied Science **New York, NY**
M.S. in Computer Science: Thesis Track (GPA: 4.00/4.33) Sep. 2020 - May 2022
- Thesis: *Statistical and Computational Methods for Neural Data Analysis*
- Advisors: Prof. Liam Paninski, Prof. David Knowles

Columbia University, School of Engineering and Applied Science **New York, NY**
B.S. in Computer Science: Intelligent Systems Track (GPA: 3.91/4.33) Sep. 2013 - May 2019
- Leave of absence for military service, Republic of Korea (Nov. 2014 - Aug. 2016)

RESEARCH EXPERIENCE

Stanford University, Computer Science Department & Wu Tsai Neurosciences Institute **Stanford, CA**
Ph.D. Student, Prof. Scott Linderman, Prof. Emily Fox Sep. 2022 - Present
- Developed Echo (under review), which teaches language models to reason in latent space by self-distilling their own chain-of-thought, outperforming prior latent methods and running far faster than explicit reasoning
- Co-identified the "curse of ambiguity" (NeurIPS 2026 spotlight), showing in theory and practice that language models need more capacity, larger embeddings, longer training, and more data to learn ambiguous next-token distributions
- Designed Switching Linear Attention (COLM 2026), a recurrent language model architecture based on mixtures of linear regressions for improved associative recall capabilities
- Developed Stiefel manifold dynamical systems (Neural Computation 2026), time-varying state-space models that track representational drift with a smoothly changing mapping from latent dynamics to neurons
- Co-developed a unifying framework (TMLR 2026) that casts fixed-point methods for parallelizing sequential models as linear dynamical systems and characterizes when each converges quickly
- Introduced switching autoregressive low-rank tensor models (NeurIPS 2023), probabilistic models that capture long-timescale dependencies with fewer parameters while keeping inference exact and simple

Columbia University, Center for Theoretical Neuroscience **New York, NY**
Graduate Research Assistant, Paninski Lab, Prof. Liam Paninski Sep. 2020 - May 2022
- Developed decentralized point-cloud-based registration (NeurIPS 2021) applied to each spike's 3D location and brightness, inferred with a point-neuron model, to track probe drift more accurately than traditional template-based methods
- Used multi-task neural networks to measure how much neural activity is explained by task, behavior, and lab-specific variables (eLife 2025), identifying factors that harm reproducibility across labs

Seoul National University, School of Computer Science and Engineering **Seoul, South Korea**
Research Staff, Data Mining Lab, Prof. U Kang Jul. 2019 - Jul. 2020
- Developed AUBER (PLOS ONE 2021), a method that leverages reinforcement learning to regularize BERT by automatically finding the optimal pruning policy
- Co-developed FALCON (Knowledge and Information Systems 2023) by defining an extended Hadamard product approximation of convolution kernels

Seoul National University, School of Computer Science and Engineering **Seoul, South Korea**
Summer Research Intern, Biointelligence Lab, Prof. Byoung-Tak Zhang Jun. 2018 - Aug. 2018
- Built a multi-label video classification network (ECCV Workshops 2018) that predicts new labels from the video and previously observed labels, with a team that placed 5th in the YouTube-8M Kaggle challenge

AWARDS & HONORS

Stanford Interdisciplinary Graduate Fellow, Stanford University 2024
NeuroTech Trainee, Stanford University 2024
Magna Cum Laude, Columbia University 2019
Dean's List, Columbia University 2013 - 2019
Tau Beta Pi Engineering Honor Society, Columbia University 2017

PUBLICATIONS

*equal contribution

Zucchet, N., **Lee, H.**, Linderman, S. (2026). Language models suffer from a curse of ambiguity. *Advances in Neural Information Processing Systems (Spotlight)*.

Lee, H., Gonzalez, X., Zucchet, N., Buchanan, E.K., Fox, E., Linderman, S. (2026). Switching Linear Attention. *Conference on Language Modeling*.

Lee, H., Jha, A., Clarke, S., Silvernagel, M., Nuyujukian, P., Linderman, S. (2026). Stiefel Manifold Dynamical Systems for Tracking Representational Drift. *Neural Computation*.

Gonzalez, X.*, Buchanan, E.K.*, **Lee, H.**, Liu, J.W., Wang, K.A., Zoltowski, D.M., Kozachkov, L., Ré, C., Linderman, S. (2026). A Unifying Framework for Parallelizing Sequential Models with Linear Dynamical Systems. *Transactions on Machine Learning Research*.

International Brain Laboratory, et al. (2025). Reproducibility of in vivo electrophysiological measurements in mice. *eLife*, 13, RP100840.

Lee, H., Warrington, A., Glaser, J., Linderman, S. (2023). Switching Autoregressive Low-rank Tensor Models. *Advances in Neural Information Processing Systems*.

Jang, J.*, Quan, C.*, **Lee, H.**, Kang, U (2023). FALCON: lightweight and accurate convolution based on depthwise separable convolution. *Knowledge and Information Systems: 65(5),* 2225-2249.

Boussard, J.*, Varol, E.*, **Lee, H.**, Dethe, N., Paninski, L. (2021). Three-dimensional spike localization and improved motion correction for Neuropixels recordings. *Advances in Neural Information Processing Systems*.

Lee, H.*, Lee, S.*, Kang, U (2021). AUBER: Automated BERT Regularization. *PLOS ONE 16.6: e0253241.*

Kim, E., On, K., Kim, J., Heo, Y., Choi, S., **Lee, H.**, Zhang, B. (2018). Temporal Attention Mechanism with Conditional Inference for Large-Scale Multi-Label Video Classification. *Proceedings of the ECCV Workshops*.

PREPRINTS

*equal contribution

Willett, F., Li, J., Le, T., Fan, C., Chen, M., Shlizerman, E., Chen, Y., Zheng, X., Okubo, T., Benster, T., **Lee, H.**, Kounga, M., Buchanan, E.K., Zoltowski, D.M., Linderman, S., Henderson, J. (2024). Brain-to-Text Benchmark '24: Lessons Learned. *arXiv*:2412.17227.

Boussard, J.*, Windolf, C.*, Hurwitz, C.*, **Lee, H.**, Yu, H., Winter, O., Paninski, L. (2023). DARTsort: A modular drift tracking spike sorter for high-density multi-electrode probes. *bioRxiv*.

TECHNICAL REPORTS

International Brain Laboratory, et al. (2022). Spike sorting pipeline for the International Brain Laboratory.

CONFERENCE ABSTRACTS

*equal contribution

Lee, H., Jha, A., Clarke, S., Silvernagel, M., Nuyujukian, P., Linderman, S. (2025). Stiefel manifold dynamical system for tracking neural drift across sessions. *Computational and Systems Neuroscience*.

Lee, H., Glaser, J., Warrington, A., Susoy, V., Samuel, A., Paninski, L., Linderman, S. (2023). Switching autoregressive low-rank tensor models for neural dynamics. *Computational and Systems Neuroscience*.

Bonacchi, N., Bruijns, S., Chapuis, G., Churchland, A., Faulkner, M., **Lee, H.**, Meijer, G., Paninski, L., Roth, N., Steinmetz, N., Taheri, M., West, S., Winter, O., and International Brain Laboratory (2021). Reproducibility of electrophysiological measurements during decision-making. *Society for Neuroscience*.

PATENT

Lee, S., **Lee, H.**, and Kang, U (2022). Pruning Method for Attention Head in Transformer Neural Networks for Regularization and Apparatus Thereof. *Korea Patent No. 10-2437750*.

INVITED TALKS

Switching Linear Attention, PLS AI Journal Club, Gore Lab, MIT — Oct. 2026

ACADEMIC SERVICE & TEACHING EXPERIENCE

Reviewer for NeurIPS (2024, 2025, 2026), ICLR (2024, 2025, 2026)

Teaching Assistant, CS 229B: Machine Learning for Sequence Modeling, Stanford University — Fall 2023

Teaching Assistant, COMS 6998: Computation and the Brain, Columbia University — Fall 2021

Teaching Assistant, COMS 4721: Machine Learning for Data Science, Columbia University — Spring 2018